

Alignment, Power, and Performance:

Quantifying Rowing Synchrony and Speed in Collegiate Rowing

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Curiosity

As a student-athlete on the Rutgers University women's rowing team, I was driven by both personal passion and academic interest to investigate what makes boats go faster. My dual role fueled a *curiosity* about whether the key to more boat speed lies not just in how hard we row, but how aligned we row. This motivation stemmed from countless early mornings on the Raritan River, where I noticed that even when our crew pulled with equal power on different mornings, the boat felt faster and smoother on days when our timing was perfectly in sync. I wanted to marry my first-hand athletic experience with data analytics to answer a pressing question: **How does alignment between rowers affect boat speed, and how does this alignment moderate the effect of the average power of all rowers on a given boat on boat speed?** In other words, can perfect timing amplify the benefits of raw power on speed?

From an academic perspective, this project aligns with my fascination for sports science and data analysis. I have always been *curious* about the physics and physiology underlying elite performance. Rowing is an ideal sport to study because it's a highly technical, quantifiable team sport where subtle differences in coordination can translate into measurable outcomes like split times. The idea of using data measured by high-tech sensors to capture every nuance of our rowing strokes excited me. It promised a way to turn subjective feelings of "swing" often derived as that almost-magical sensation when a boat is moving effortlessly into hard data that could be studied and optimized. This natural curiosity set the stage for a rigorous inquiry, motivated by a desire to improve our boat's performance and contribute knowledge at the intersection of sports and data science.

On a personal level, I was inspired by the possibility that insights from this research could help my teammates and I row more effectively. The Honors College values of curiosity, knowledge, and purpose deeply resonated throughout this journey: curiosity sparked the inquiry; knowledge would be built through research; and purpose came from aiming to apply findings in a real-world team setting. With these motivations in mind, I set out to design a study that could quantitatively capture the elusive qualities of synchronization and alignment in rowing and determine their impact on boat speed.

Knowledge

Data Collection during Rutgers Women's Rowing Team 2024 Regatta Season

To transform my questions into evidence-based answers, I used a rich dataset collected during all competitive races of the Rutgers First Varsity 4 Boat, i.e. a rowboat with 4 rowers, during the Spring 2024 regatta season. Each of the 11 races in our season became not just an athletic contest but also a data-gathering opportunity. Our boat was equipped with Nielsen-Kellerman Empower Oarlock sensors on each rower's oar. These high-tech oarlocks recorded detailed per-stroke metrics for every rower in the boat which provided a granular view of our performance.

Key per-stroke data collected from Oarlocks and boat speed coach include:

- *Power output:* The power (in watts) each rower produced on each stroke. This per-stroke power data reflects how hard the rower was pulling.
- *Catch and finish angles and stroke length:* The oar angle at the catch and finish for each stroke. These angles indicate how far towards the bow, i.e. towards the front to the boat

(catch angle) and towards the stern, i.e. towards the back to the boat (finish angle) each rower's oar traveled. They measure the length of each rower's stroke. Differences in catch and finish angles between rowers can reveal mismatches in timing or technique.

- *Wash and Slip*: Slip measures the amount of motion in degrees between when the blade enters the water (catch) and when over 200 Newtons of force are being produced on the blade. Wash measures the amount of motion in degrees between when the force on the blade stops producing over 200 Newtons and the blade exits the water (finish).
- *Max Force Angle*: Angle (in degrees) of when there is the most amount of force (in Newtons) against the blade.
- *Boat speed and timing*: Although not directly from the oarlock, the **boat's speed for each stroke** (meters per second via the boat's speed coach GPS) and the **stroke rate** (strokes per minute) were recorded in parallel. This is crucial for correlating rowers' input (power and timing) with output (boat speed).

Wind data collected include:

In rowing, wind is one of the most influential environmental factors affecting boat speed. It can either aid or hinder performance depending on its direction and strength relative to the racecourse. A tailwind increases boat speed, while a headwind makes boats slower. That's why incorporating wind velocity data (speed and direction) for each race is essential when modeling or comparing rowing performance.

- *Wind velocity*: Historical wind direction and speed data sourced from the nearest weather station to the racecourse from wunderground.com.

Data Cleaning and Feature Engineering

Raw data in hand, the next step was to transform it into a clean, analyzable format. This involved rigorous data cleaning followed by feature engineering to extract meaningful variables that quantify technical alignment between rowers and performance. The stroke-level data from the Empower Oarlock sensors were originally stored separately for each rower and race, requiring a comprehensive merging process. To reconstruct full-boat dynamics, datasets were merged on a per-stroke, per-race basis using stroke number alignment and a unique race identifier. This allowed rower-level metrics (power, angle, length) to be aggregated and compared across all four crew members within the same boat for each stroke.

Other key cleaning steps included:

Removing Duplicates: Duplicate entries caused by sensor buffering or overlapping logs were identified and eliminated using a unique composite key of race ID, rower ID, and stroke number.

Filtering by Stroke Range: Only strokes numbered between 100 and 200 were retained to focus on the middle segment of each race, which is typically the most stable and least affected by start-line acceleration or end-of-race sprint.

Ensuring Numeric Integrity: All relevant fields (power, angles, rates) were converted into numeric format.

Feature engineering:

With clean data, I moved to creating new variables that could quantify rower alignment and other complex factors in a meaningful way:

- *Wind Along and Wind Cross*: Recognizing the large influence of environmental factors, I created two wind-related features:
 - **Wind Along**: the component of wind aligned with the boat's travel direction (headwind or tailwind in miles/hour).
 - **Wind Cross**: the component perpendicular to the boat's travel direction (crosswind in miles/hour).

To calculate these, I determined the heading angle of each racecourse from GPS start and finish coordinates and combined this with historical wind direction and speed data sourced from the nearest weather station located near the racecourse from wunderground.com. The wind vector was decomposed into its along-course and across-course components using basic trigonometry:

- $Wind\ Along = Wind\ Speed \times \cos(\theta)$

- $Wind\ Cross = Wind\ Speed \times \sin(\theta)$

■ where θ is the angle between the boat's heading and the wind direction.

- *Average Power per stroke*: For each boat and stroke, I computed the *Average Power per Stroke* as the mean power output of all four rowers for each stroke in watts.
- *Effective Length*: To compute the most powerful part of each stroke, i.e. the part of each stroke that propels the boat with a power of 200 Newtons or more, I calculated each rower's *Effective Length* per stroke by combining sensor-measured catch and slip angles at the front end of the stroke and finish and wash angles at the back end:
 - **Effective Length**: $|(Catch + Slip) - (Finish - Wash)|$

- *Effective Catch and Effective Finish:* To more accurately capture entry and exit positions of the oar that translate into greater power (>200 Newtons), I constructed two additional stroke-level variables for each rower:
 - **Effective Catch** = Catch + Slip
 - **Effective Finish** = Finish - Wash

These metrics integrated slip and wash (measures of missed water at catch and finish, respectively) with the corresponding oar angles to describe the range of movement of the oars in the water that contributes most to boat speed

- *Boat-Level Technical Alignment Measures:* To measure technical alignment of the four rowers on the boat, I created variables that measure the variability of power and stroke mechanics between the rowers with lower variability indicating a greater alignment between rowers. To do so, I aggregated per-stroke data at the boat level (grouped by race and stroke number) and calculated the following variables for each stroke across the four rowers on the boat:
 - **Standard Deviation of Catch Angles (SD_Catch)**
 - **Standard Deviation of Finish Angles (SD_Finish)**
 - **Standard Deviation of Effective Catch (SD_Effective_Catch)**
 - **Standard Deviation of Effective Finish (SD_Effective_Finish)**

A lower standard deviation indicates lower variability and thus greater technical alignment among rowers.

- *Boat Level Power Alignment Measures:* For each boat and stroke, I computed:

- **Standard Deviation of Power (SD_Power):** Variability in rowers' power application (in Watts) within the boat for each stroke. A lower SD_Power implies a more balanced collective effort and thus greater alignments, while higher variability suggests mismatches in strength.
- *Standardized Features and Interaction Terms:* To test whether alignment between rowers moderates the relationship between the overall effort, i.e. the average power across rowers for each stroke, and boat speed I created interaction terms between standardized measures of (mis)alignment between rowers and standardized average power. I standardized (zero-mean, unit-variance scaling) the following variables:
 - **Avg Power z-score:** Standardized average power.
 - **SD_Power z-score:** Standardized power variability.
 - **SD_Effective_Catch z-score:** Standardized effective stroke catch variability
 - **SD_Effective_Finish z-score:** Standardized effective strokefinish variability

I created three interaction terms that allow modeling whether the effect of adding greater average more power on boat speed depended on the level of technical (mis)alignment between the rowers on the boat. Interaction terms were created by multiplying the standardized average power with each variability metric:

- **PowerFit_Interaction1_z:** Avg Power $\times$ SD Power z-score
- **PowerFit_Interaction2_z:** Avg Power $\times$ SD Effective Catch z-score
- **PowerFit_Interaction3_z:** Avg Power $\times$ SD Effective Finish z-score

Modeling Approach and Key Results

With a refined dataset in place, I proceeded to the analytical core of the project: modeling the factors influencing boat speed. I took a two-pronged modeling approach to capture both non-linear interactions and hierarchical data structure.

Random Forest Regression

I first employed a random forest model to explore which factors were most important in predicting boat speed. A random forest is an ensemble of decision trees and is great for detecting complex non-linear relationships and interactions without needing to explicitly pre-specify them. I fed the model a wide array of features including average power per stroke, stroke rate, catch angle, wind conditions, etc. and allowed it to determine the importance of each.

The random forest results were illuminating (See Appendix 1). The model's variable importance metrics indicated that wind conditions stood out as the most significant factor affecting boat speed. This confirmed quantitatively what rowers often feel qualitatively: a strong headwind can slow the boat markedly, and a tailwind can give a helpful push. In fact, the wind-related features (wind speed and whether it was head or tail) had the highest importance scores.

Among the rower-generated factors, average power per stroke was obviously crucial as more power generally led to higher speeds. Additionally, stroke rate (how fast the crew was rowing) was important, as expected. The random forest provided a broad, data-driven confirmation that power was indeed a key factor, alongside the uncontrollable factor of wind.

Mixed-Effects Linear Modeling

To dive deeper and quantify the relationships between rowing mechanics and boat speed, I used a linear mixed-effects model (See Appendix 2). This model was appropriate because the dataset had a hierarchical structure: strokes nested within races, with different crew lineups. By treating each race as a random effect, the model could account for unobserved race-level factors, such as overall crew ability or environmental conditions (e.g., choppy water vs. calm water), that were not directly measured.

In this model, the dependent variable was boat speed per stroke (measured by GPS). The fixed effects included:

- Average Power per Stroke
- Stroke Rate
- Effective Length
- Max Force Angle
- Standard Deviation of Power
- PowerFit_Interaction1_z
- Standard Deviation of Effective Catch
- Standard Deviation of Effective Finish

Average power per stroke had a statistically significant and positive coefficient ($p < 0.05$). As expected, more power led to higher boat speeds. This affirms the intuitive principle that if the crew as a whole is putting more watts into the water, the boat should move faster. This means for example in a 2,000 meter race if a crew pulled on average 20 more watts, they can expect a 0.02 m/s speed increase per stroke which translates to an about 2 second decrease in overall time,

which could be the difference between winning or losing a race. $\text{Power} \times \text{SD_Power}$ ($\text{PowerFit_Interaction1_z}$) was statistically significant with a negative coefficient. This negative interaction means that the positive effect of average power on boat speed is diminished when synchronization of power across the four rowers on the boat is poor. In contrast, when synchronization is high, the effect of power on speed is stronger. This finding confirms my intuition: alignment moderates the relationship between power and speed. Two crews applying the same average power can see different boat speeds depending on how well they row in unison.

Stroke Rate had a small positive coefficient, but it was not statistically significant, suggesting that simply increasing rate, without improving timing or power, may not meaningfully improve speed. Effective Length and Max Force Angle both had negative and significant coefficients, indicating that greater values in these metrics were associated with slightly lower speeds. Variability in catch and finish angles ($\text{SD_Effective_Catch}$ and $\text{SD_Effective_Finish}$) were not statistically significant, although both trended slightly negatively, implying that misalignment in stroke endpoints may have a small detrimental effect, but not conclusively. This mixed effects model was also done with the other two interaction terms, but neither of these interactions proved significant.

Implications of Findings

The mixed-effects model provides strong statistical support for the idea that synchronization enhances the effectiveness of power output. While raw power is a key determinant of boat speed, it is not sufficient alone, its impact is significantly amplified when the rowers are well-synchronized. This highlights an important coaching takeaway: maximizing boat

speed requires not just powerful strokes, but also precise timing across all rowers. Even small misalignments in timing can blunt the gains from added power.

The combined evidence from the random forest and mixed-effects models provided strong, evidence-based conclusions. Firstly, boat speed is a multifactor outcome. While power and rate are fundamental, environmental factors like wind can heavily influence results and must be accounted for. Secondly, rower alignment and synchronization, while intangible to the naked eye beyond “that looks smoother,” can be quantified and do make a difference in performance. Being in sync isn’t just aesthetically pleasing; it has a measurable effect on how effectively power translates into speed. Lastly, The interplay between raw power and technique is crucial. Teams that can harness both will outperform those that have one without the other. This reinforces long-held coaching wisdom with data: “Four strong individuals won’t beat a weaker but perfectly synchronized crew” up to a point as the data now puts numbers to that assertion .

Academic Preparation and Analytical Skills

Executing this study successfully required not just rowing know-how but also a solid grounding in empirical research methods and statistics. Thankfully, my academic preparation, especially the graduate-level courses in *Experimental Design* and *Multivariate Analysis*, proved invaluable at each step.

In *Experimental Design*, I learned about structuring studies to test hypotheses while controlling for variability. I applied these principles by treating each race almost like a blocked experiment and each stroke as a repeated measure. The concept of blocking by race and recognizing when to include random effects stem directly from experimental design thinking.

Meanwhile, *Multivariate Analysis* introduced me to handling multiple variables and their interrelationships. This was crucial when interpreting the random forest output and in constructing the mixed model. With many features, I had to ensure that highly collinear features did not distort the model. The course taught me to check variance inflation factors and to be cautious about including closely related variables (e.g., average power and total work done per stroke are proportional given a fixed stroke rate). I ended up creating composite features or dropping redundant ones based on these diagnostics. Multivariate thinking also guided me to test interactions like power*synchronization.

Throughout the *Knowledge* phase of this project, I practiced scientific thinking: formulating a question, gathering data meticulously, analyzing it with appropriate methods, and then critically interpreting what the data meant. Each finding was cross-checked or visualized to ensure it made sense. For instance, after seeing the model results, I plotted boat speed vs. the synchronization index of each race to visually confirm the slopes. The data points during highly synchronized times showed a steeper slope (more speed gain per power) than during less synchronized times. Such evidence-based reasoning was key to solidifying the conclusions.

In summary, this project expanded my knowledge in several dimensions. Technically, I gained hands-on experience with data logging technology and advanced statistical modeling. In terms of rowing science, I used one of the most detailed datasets at the collegiate level, and extracted insights that can help coaches and athletes. And intellectually, I learned how to connect the dots between theory, empirical data, and practical implications; a skill that lies at the heart of scholarly research.

Purpose

Reflecting on this journey, I realize that this project was about far more than just numbers and rowing speeds. It was a personal and scholarly quest that taught me profound lessons about teamwork, consistency, and the very nature of pursuing knowledge with a purpose.

Lessons on Teamwork

One immediate takeaway was how interconnected every member of a rowing crew is. Even though my analysis treated each stroke as a data point, behind each data point were five humans (four rowers and a coxswain) trying to move in harmony. Through the variations in the rowboat data, I saw a reflection of our teamwork. I learned that teamwork isn't just a feel-good concept; it has a tangible effect on performance metrics. This insight will stay with me in any collaborative endeavor: success comes when a group operates with shared purpose and timing, whether in sports, in a lab, or in a workplace team project.

Lessons on Technical Consistency

Rowing races are won by consistent execution, and similarly, data-driven research succeeds through consistent methodology. Balancing being a student athlete and a student researcher was challenging and helped me become more disciplined. This discipline in approach reflects a larger lesson: in any rigorous undertaking, consistency and attention to detail are paramount.

Challenges of Rigorous Sports Science Research

Conducting this study gave me a first-hand appreciation of the challenges inherent in sports science. Unlike a controlled lab experiment, a rowing race is a whirlwind of

uncontrollable variables. Weather can change mid-race, a competitor's move can force a strategy shift, or equipment can fail. I faced the real-world messiness of data: I had to account for things like the coxswain's calls (which might prompt a surge in power or boatspeed), varying adrenaline levels, and even human error in rowing technique. Designing a study that holds up to academic scrutiny in such an environment taught me creativity and resilience. I realized the importance of sample size and repeated measures; a single race or two might have misleading patterns, but across 11 races, true signals emerged that superseded the noise. This reinforced a fundamental research principle: repeat and validate findings across conditions to ensure they're robust. It was challenging to find the right balance between capturing reality and controlling for enough variables to make analysis feasible, but it was incredibly rewarding to see it come together.

Guiding Future Pursuits

The capstone project has been a formative experience that I will carry into my future academic and professional life. Firstly, it confirmed my interest in the field of sports science and data analytics. I discovered the joy of using data to crack open performance puzzles. In the future, I could see myself pursuing graduate studies in Operations Research. The blend of engineering, data science, and human physiology in this project was thrilling, and it's exactly the kind of interdisciplinary challenge I want to tackle in my career.

Finally, the personal growth I experienced has set a tone for how I will approach challenges going forward. I gained confidence that I can identify a problem in an area I'm passionate about, apply scholarly methods to investigate it, and derive insights that have practical value. This is a powerful realization as I step out of college: it means that whenever I encounter a

problem, be it improving a process at work or understanding a societal issue, I have the toolkit to dive in systematically and the motivation to follow it through with purpose.

In conclusion, this Honors College Capstone project on rowing performance was a microcosm of what it means to engage deeply with curiosity, build and apply knowledge, and pursue a meaningful purpose. It has reinforced my love for learning and discovery and has set me on a path of lifelong inquiry. Just as importantly, it has shown me how knowledge gains greater significance when it can be applied to something one cares about much like how a boat moves fastest when every rower is pulling in unison with power and precision. I am excited to carry this blend of intellectual rigor and passionate purpose into the next chapter of my journey, wherever it may lead.

Appendix 1:

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Random Forest Feature Importances:
Wind_Along      0.521504
Wind_Cross      0.175310
Force Avg       0.042251
Work            0.039829
Max Force Angle 0.037424
Power           0.036432
Finish          0.031572
Slip            0.025760
Catch           0.025149
Effective Length 0.023964
Wash            0.020597
Stroke Rate     0.020209

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Appendix 2:

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=== Mixed Model: All Races ===
      Mixed Linear Model Regression Results
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Model:             MixedLM   Dependent Variable:  Q('Speed (GPS)')
No. Observations:  3312      Method:              REML
No. Groups:         11       Scale:              0.0088
Min. group size:    200      Log-Likelihood:   3055.4421
Max. group size:    404      Converged:        Yes
Mean group size:    301.1

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              Coef.  Std.Err.  z      P>|z|  [0.025 0.975]
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Intercept          4.294    0.093  46.295  0.000   4.112   4.476
Q('Stroke Rate')    0.003    0.002   1.362  0.173  -0.001   0.006
Q('Effective Length') -0.004    0.001  -7.633  0.000  -0.005  -0.003
Q('Max Force Angle') -0.001    0.000  -4.940  0.000  -0.001  -0.001
PowerFit_Interaction1_z -0.004    0.001  -6.029  0.000  -0.005  -0.003
SD_Power            0.001    0.000   3.493  0.000   0.000   0.001
SD_Effective_Catch  -0.001    0.001  -0.723  0.470  -0.003   0.001
SD_Effective_Finish -0.002    0.002  -1.131  0.258  -0.006   0.002
Avg_Power_per_Stroke 0.001    0.000   5.311  0.000   0.001   0.002
Group Var           0.023    0.109

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